“HOTEL MANAGEMENT SYSTEM”

Name: KHAN SHAGUFTA/KHAN IQRA

Class/Batch: SYIT

Project Title: HOTEL MANAGEMENT SYSTEM USING PYTHON USING GUI

1. Project Description:

Hotel Management System is a Python-based GUI application designed to simplify and automate daily hotel operations. The system provides an easy-to-use graphical interface for managing **room bookings, customer details, check-in/check-out, room availability, billing, and payment records**. It reduces manual work, improves accuracy, and helps hotel staff manage information efficiently.

This code demonstrates the main functionality of the Hotel Management System using **Python Tkinter GUI**. It allows users to select a room, enter check-in/check-out dates, calculate the number of days, and automatically generate the total bill based on the room price.

2. Code Snippet:

```
import tkinter as tk

from tkinter import ttk, messagebox

from datetime import datetime


root = tk.Tk()

root.title("Hotel Management System")

root.geometry("1050x650")


# Room prices
```

```
room_prices = {
    "101 - Single Room": 1500,
    "102 - Single Room": 1500,
    "201 - Double Room": 2500,
    "301 - Deluxe Room": 3500,
    "401 - Suite": 5000
}

def calculate_bill():
    try:
        checkin = datetime.strptime(checkin_var.get(), "%d-%m-%Y")
        checkout = datetime.strptime(checkout_var.get(), "%d-%m-%Y")

        if checkout <= checkin:
            messagebox.showerror("Error",
                                 "Check-out must be after check-in")
            return

        days = (checkout - checkin).days
        price = room_prices[room_var.get()]
        total = days * price

        days_var.set(str(days))
        amount_var.set(f"Rs. {total}")

    except ValueError:
```

```
        messagebox.showerror(
            "Error", "Enter date in DD-MM-YYYY format"
        )

# GUI variables
name_var = tk.StringVar()
phone_var = tk.StringVar()
room_var = tk.StringVar()
checkin_var = tk.StringVar()
checkout_var = tk.StringVar()
days_var = tk.StringVar()
amount_var = tk.StringVar()

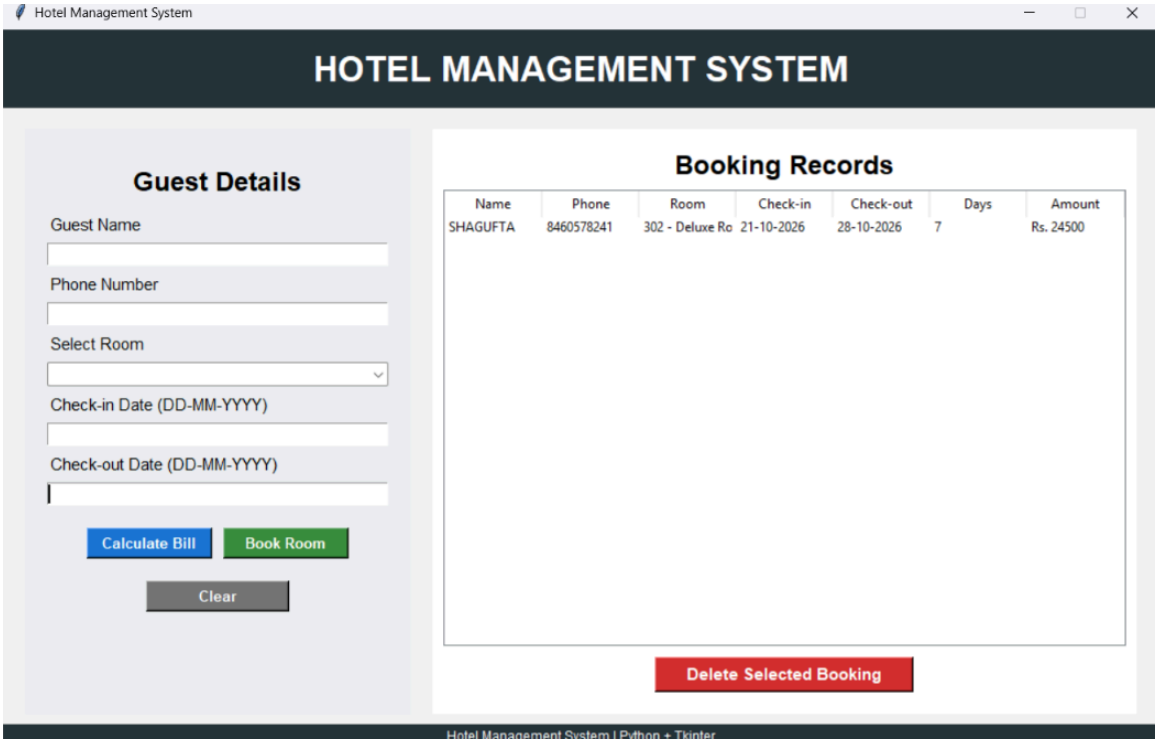
# Room selection
room_box = ttk.Combobox(
    root,
    textvariable=room_var,
    values=list(room_prices.keys()),
    state="readonly"
)
room_box.pack(pady=10)

# Calculate bill button
tk.Button(
    root,
    text="Calculate Bill",
```

```
    command=calculate_bill
).pack(pady=10)

root.mainloop()
```

3. Output Screenshot:



6. Additional Notes:

Challenges Faced

- Designing a user-friendly GUI using **Tkinter**.
- Validating check-in and check-out dates correctly.
- Calculating the total bill based on room price and number of days.
- Preventing duplicate room bookings.

- Managing booking records using the **Treeview** widget.
- Handling invalid or missing user inputs using error messages.

Future Improvements

- Add a **database** such as MySQL or SQLite to permanently store bookings.
- Add **login and user authentication** for hotel staff.
- Add online payment and invoice generation.
- Add automatic room availability/status management.
- Add customer search, update, and booking history features.
- Improve the GUI with modern themes, icons, and better navigation.

Learning Outcomes

- Learned how to develop a GUI application using **Python Tkinter**.
- Understood event-driven programming and functions.
- Learned input validation and error handling.
- Gained experience with **date calculations and billing logic**.
- Learned how to display and manage records using Tkinter's **Treeview**.
- Improved problem-solving, debugging, and Python programming skills.